PM-04-PS05

Program Basic Algorithms and Evaluate Program Correctness

Robotic Process Automation Solution

https://github.com/TheFish1011/PM-04-PS05.git

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Course Code	PM-04-PS05
Document Version	v1.0
Submission Date	27 April 2026

1. Worksheet

1. Map out the automation workflow
PM-04-PS05/Workflow_Map_PM04-PS05.md at main · TheFish1011/PM-04-PS05

2. Detailed build guide
PM-04-PS05/Data/Resources/Build_Guide_PM04-PS05.txt at main · TheFish1011/PM-04-PS05

3. Testing Evidence
Refer to TestExtractFromPDF.xaml

Refer to TestWriteOutputCSVOnly.xaml

These two workflows does not form part of the main and was built only to test extraction. Various Log Messages are also included in all workflows so that logs can give indiciation of where errors occur.

Refer to Data/Output

CSV_Test_Results.xls

PDF_Test_Results.xls

Are outputs generated by above workflows to ensure data is correct.

4. Error Logs

Refer to Data/Resources

Log_PM04_PS05.txt provides example of log generated with no errors

5. Reflection

1. Faster Response Times

Manually sorting through hundreds of complaints submitted via email, PDF, or spreadsheets is time-consuming. This bot processes all incoming complaints in minutes, meaning issues like burst pipes or power outages get routed and logged immediately — rather than sitting in an inbox for days.

2. Consistent & Unbiased Classification

Human clerks may classify complaints differently depending on experience, workload, or interpretation. The bot applies the same rules every single time, ensuring every Water, Electricity, and Refuse complaint is categorised consistently — removing human error and subjectivity.

3. Data-Driven Decision Making

With classified complaints written to Excel and summarised in a daily email, supervisors gain instant visibility into complaint trends — e.g. a spike in Water complaints in a specific week could signal a systemic infrastructure problem that needs urgent attention.

4. Freeing Up Staff for Higher-Value Work

Municipal staff currently doing manual data entry, sorting, and reporting can be redirected to citizen-facing or problem-solving roles — improving both staff satisfaction and the quality of service delivered to the public.

5. Audit Trail & Accountability

Every complaint is logged, timestamped, and stored with an error record if something goes wrong. This creates a transparent paper trail that municipalities can use for audits, performance reviews, and compliance reporting.

6. Foundation for Smarter Automation

This rule-based bot is a solid starting point. In the future, the keyword classifier could be replaced or augmented with an AI/ML model trained on historical complaints — enabling the bot to handle more nuanced language, detect priority levels, and even predict complaint volumes by area or season.